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Prediction of Thermodynamic and Electronic Molecular Properties Using Functional Group-Based Machine Learning Models

Data Science in Chemical Engineering

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Abstract: Accurate prediction of molecular thermodynamic and electronic properties is essential in chemical engineering, materials science, and computational chemistry. While quantum chemical methods provide reliable estimates, their computational cost limits large-scale use. This work develops a data-driven framework based on the QM9 dataset to predict twelve molecular properties from chemically interpretable descriptors. The workflow starts from raw `.xyz` files, extracts SMILES representations, and generates feature sets based on functional group counts, atomic composition, and physicochemical descriptors computed with RDKit. Classical machine learning models—Ridge Regression, Random Forest, and Histogram Gradient Boosting—are compared with feedforward neural networks under two scenarios: before and after feature engineering. Feature engineering strongly improves performance, increasing average R^2 from about 0.75 to about 0.91. The largest gains are observed for shape- and vibration-sensitive properties such as r^2, C_v, and the HOMO–LUMO gap. Classical models reach accuracy comparable to neural networks while preserving higher interpretability through SHAP analysis. Dipole moment remains the most challenging target because two-dimensional descriptors do not fully represent three-dimensional geometry and electronic asymmetry. The results show that domain-informed descriptors provide an effective and scalable alternative to direct quantum chemical calculations for molecular property screening.

Key-words: molecular property prediction, QM9, functional groups, feature engineering, machine learning, neural networks

1. Introduction

Prediction of molecular properties is a central task in chemical engineering, materials design, and quantum chemistry. Electronic and thermodynamic quantities such as orbital energies, dipole moment, polarizability, and heat capacity govern molecular behavior and are therefore crucial for process design and screening. Quantum chemical calculations can provide these properties with high accuracy, but their computational cost remains a major limitation when large molecular libraries must be explored.

An alternative route is to learn structure–property relations directly from data. The conceptual basis of this idea can be traced back to group contribution methods, where molecular properties are approximated from the additive influence of functional groups [1]. Modern machine learning extends this principle by learning

nonlinear relations from large datasets such as QM9, which contains quantum-mechanical properties for more than 130,000 small organic molecules [4].
Recent literature confirms the value of chemically informed representations. Jung et al. combined substructure embeddings with feature selection to improve quantum property prediction [2]. Zeng et al. showed that image-based self-supervised learning can also recover chemically meaningful representations [5]. ChemXploreML further highlighted the effectiveness of functional group descriptors and classical machine learning for molecular prediction tasks [3]. Motivated by these studies, the present work focuses on interpretable features derived from functional groups and elemental composition, and evaluates their impact on both classical machine learning and neural network models.
The main objective is to build an accurate yet interpretable framework for predicting twelve molecular properties from two-dimensional structural information. Special attention is given to the effect of feature engineering and to the comparative behavior of classical and neural models.
The main contributions of this work are:

- A chemically interpretable feature engineering framework combining functional group descriptors, atomic composition, and physicochemical properties;
- A systematic comparison between classical machine learning models and neural networks across twelve quantum molecular properties;
- A detailed interpretability analysis using SHAP values to link predictive performance with chemical intuition.

2. Methodology

2.1. Dataset and Target Properties

The starting point is the QM9 dataset [4]. Each molecule is stored as an `.xyz` file containing atom coordinates, quantum properties, vibrational information, SMILES strings, and InChI identifiers. From the 17 quantities reported in the original files, 12 target properties were selected for prediction: dipole moment μ, polarizability α, HOMO, LUMO, HOMO–LUMO gap, electronic spatial extent r^2, zero-point vibrational energy (ZPVE), internal energies U_0 and U, enthalpy H, free energy G, and heat capacity C_v.
These targets cover electronic, thermodynamic, and vibrational behavior. Rotational constants and indexing metadata were excluded because they were outside the scope of the study.

2.2. From Raw `.xyz` Files to a Structured Dataset

The preprocessing pipeline converts the raw QM9 archive into a machine learning-ready table. Files are read sequentially from the compressed archive to avoid full extraction. For each valid molecule, the parser reads the second line of the `.xyz` file, extracts the required numerical targets, and retrieves the canonical SMILES string from the second-to-last line. The resulting structured record has the form

$$\{\text{index, SMILES, } \mu, \alpha, \text{HOMO, LUMO, gap, } r^2, \text{ZPVE, } U_0, U, H, G, C_v\}.$$

Robust exception handling is used to skip malformed entries or corrupted files without interrupting the full pipeline.
The distribution of target variables was analyzed to identify potential challenges for regression modeling. As shown in Figure 1, most thermodynamic and electronic properties exhibit relatively symmetric distributions. However, the dipole moment (μ) and the electronic spatial extent (r^2) display strong positive skewness, characterized by long tails toward higher values.

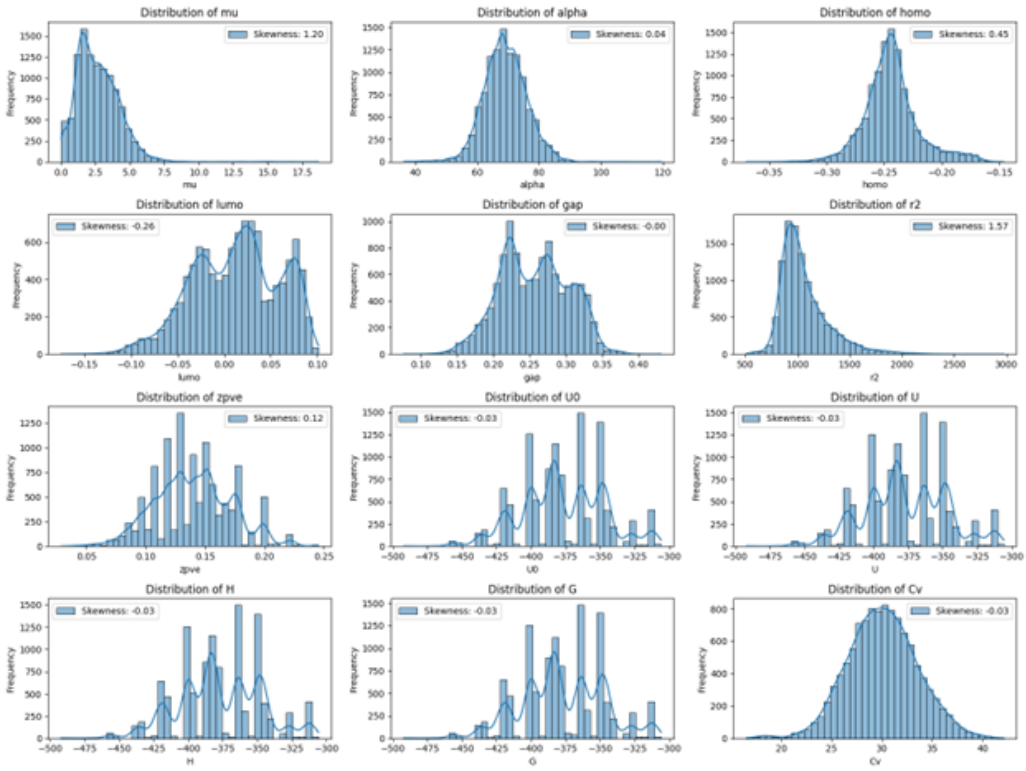


Figure 1: Distribution of target properties

2.3. Feature Generation from Molecular Structure

The modeling framework uses two successive feature sets. The baseline feature space contains simple structural descriptors derived from molecular representation. The enriched feature space adds domain-informed descriptors extracted with RDKit.

The first group of engineered features consists of functional group counts derived from SMARTS matching using RDKit's fragment catalog. These descriptors encode chemically meaningful motifs such as carbonyls, nitriles, amines, and oxygenated groups. The second group corresponds to atomic composition, represented by counts of C, H, O, N, and F atoms. A third group contains physicochemical descriptors selected for their relation to the target properties: topological polar surface area (TPSA), molar refractivity (MR), principal moments of inertia (PMI1, PMI2, PMI3), total inertia, inertia ratio, and Gasteiger charge statistics (mean and standard deviation).

This engineered feature space was designed to encode polarity, size, mass distribution, and charge heterogeneity while remaining computationally inexpensive and chemically interpretable.

2.4. Exploratory Analysis and Target Transformation

Before model training, the target variables were analyzed through histograms, boxplots, and correlation maps. The report showed strong positive skewness for dipole moment μ and electronic spatial extent r^2, while many thermodynamic targets displayed more regular distributions. Such skewed distributions can negatively impact model performance by violating the assumption of normally distributed residuals. To address this issue, a logarithmic transformation was applied to μ and r^2. As illustrated in Figure 2, this transformation reduces the influence of extreme values and produces more symmetric distributions, facilitating more stable model training.

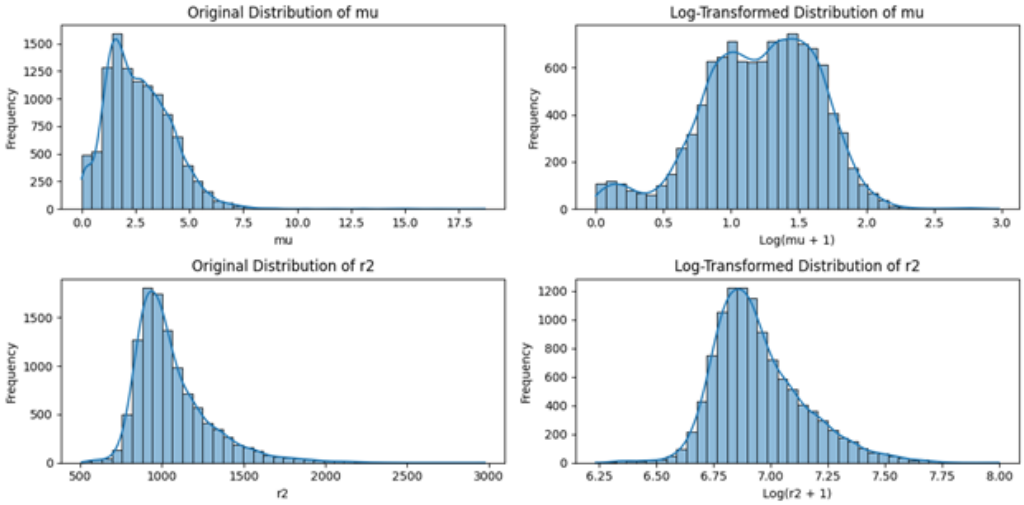


Figure 2: Effect of logarithmic transformation on μ and r^2 (see original report).

In addition, the correlation matrix shown in Figure 3 reveals strong relationships among several properties, such as polarizability (α), LUMO energy, and the HOMO–LUMO gap. In contrast, the dipole moment exhibits weak correlation with other variables, suggesting that it is governed by more complex structural and electronic factors and may therefore be more difficult to predict accurately.

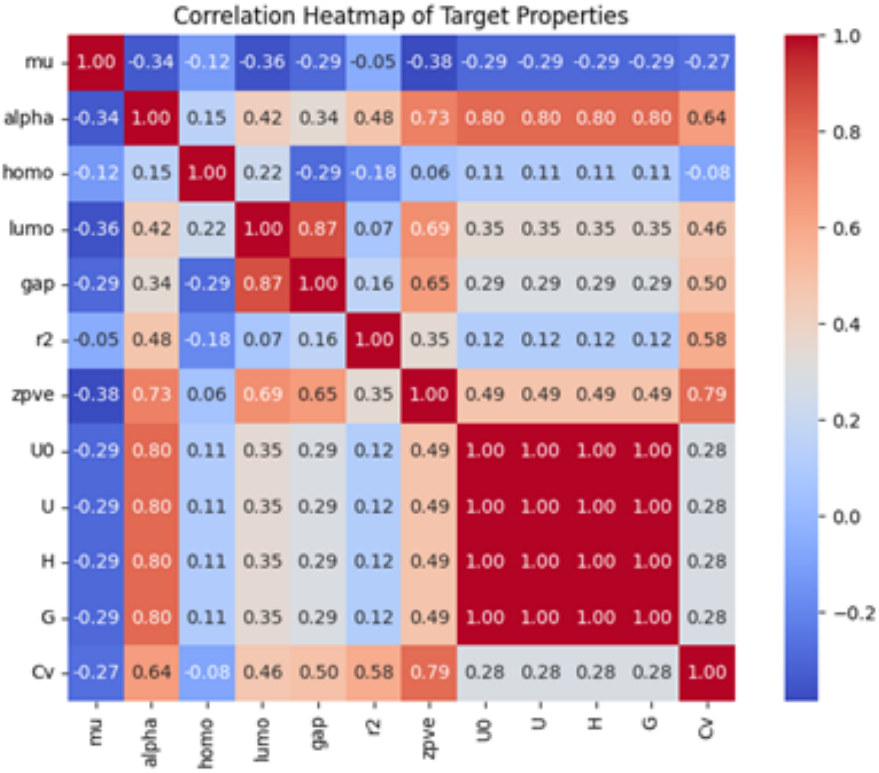


Figure 3: Correlation matrix of target properties

2.5. Modeling Strategies

The dataset was split into training and test sets using an 80/20 ratio. Model performance was evaluated using RMSE, MAE, and R^2 on the test set. All models were implemented in Python using scikit-learn and TensorFlow/Keras. Two model families were evaluated.

Classical machine learning models. Three regressors were benchmarked: Ridge Regression, Random Forest (RF), and Histogram Gradient Boosting Regressor (HGBR). Ridge provides a regularized linear baseline and is particularly suitable when targets depend mainly on additive composition. RF and HGBR capture nonlinear

effects and feature interactions, which are expected for electronic properties.
Neural networks. A feedforward neural network implemented in TensorFlow/Keras was trained for multi-target regression. One- and two-hidden-layer architectures with ReLU activation were explored. Batch normalization and dropout were used to stabilize training and mitigate overfitting. Adam was selected as optimizer and mean squared error as loss function.
For both modeling families, hyperparameter optimization followed a two-stage procedure. A broad random search identified promising regions of the search space, followed by focused refinement through grid search for classical models and Bayesian optimization for neural networks. Model performance was evaluated through RMSE, MAE, and R^2.

3. Results and Discussion

3.1. Effect of Feature Engineering

The central result of the study is the strong and systematic benefit of feature engineering. Table 1 summarizes representative performance values before and after feature enrichment. The average R^2 improved from approximately 0.75 to approximately 0.91 for both modeling families, which confirms that chemically informed descriptors are more important than model complexity alone.

Table 1: Representative prediction performance before and after feature engineering. Values are taken from the original report.

Property	Classical ML (R^2)		Neural Network (R^2)	
	Before	After	Before	After
gap	0.576	0.843	0.573	0.843
μ	0.519	0.574	0.527	0.579
α	0.839	0.963	0.842	0.963
HOMO	0.539	0.811	0.615	0.813
LUMO	0.675	0.874	0.675	0.873
r^2	0.159	0.895	0.176	0.906
ZPVE	0.678	0.975	0.686	0.970
C_v	0.481	0.921	0.495	0.930

The most dramatic gain appears for r^2, whose R^2 rises from about 0.16 to about 0.90. This indicates that descriptors related to molecular size, inertia, and charge distribution capture the physical mechanisms governing electronic spatial extent much better than the baseline representation. A similarly strong effect is observed for heat capacity C_v, which improves from about 0.48 to more than 0.92. This result is consistent with the dependence of vibrational and thermal properties on molecular flexibility, topology, and mass distribution.
The HOMO–LUMO gap, HOMO, and LUMO also benefit strongly from enriched descriptors. Their improvement suggests that properties related to frontier orbitals are not adequately represented by raw structural inputs alone and require explicit information about polarity, charge separation, and molecular geometry proxies.
For thermodynamic energies U_0, U, H, and G, performance is already excellent at baseline and remains close to unity after feature engineering. This confirms that these quantities are dominated by additive and size-related effects that can be captured well even by linear models.

3.2. Model Performance Evaluation

Model performance was further evaluated using parity plots, which compare predicted values with ground truth values. As shown in Figure 4, classical machine learning models exhibit a substantial improvement after feature engineering. Predictions become more tightly aligned with the diagonal line, indicating reduced error and improved generalization.
This improvement is particularly evident for properties such as the HOMO–LUMO gap, r^2, and heat capacity (C_v), indicating that feature engineering plays a dominant role in capturing non-linear structure–property relationships that are not adequately represented in the baseline feature space.
A similar trend is observed for neural networks in Figure 5. While the baseline model shows noticeable dispersion, the feature-engineered version produces predictions that closely follow the ideal parity line. This consistency

across both modeling approaches highlights the dominant role of feature quality over model complexity.

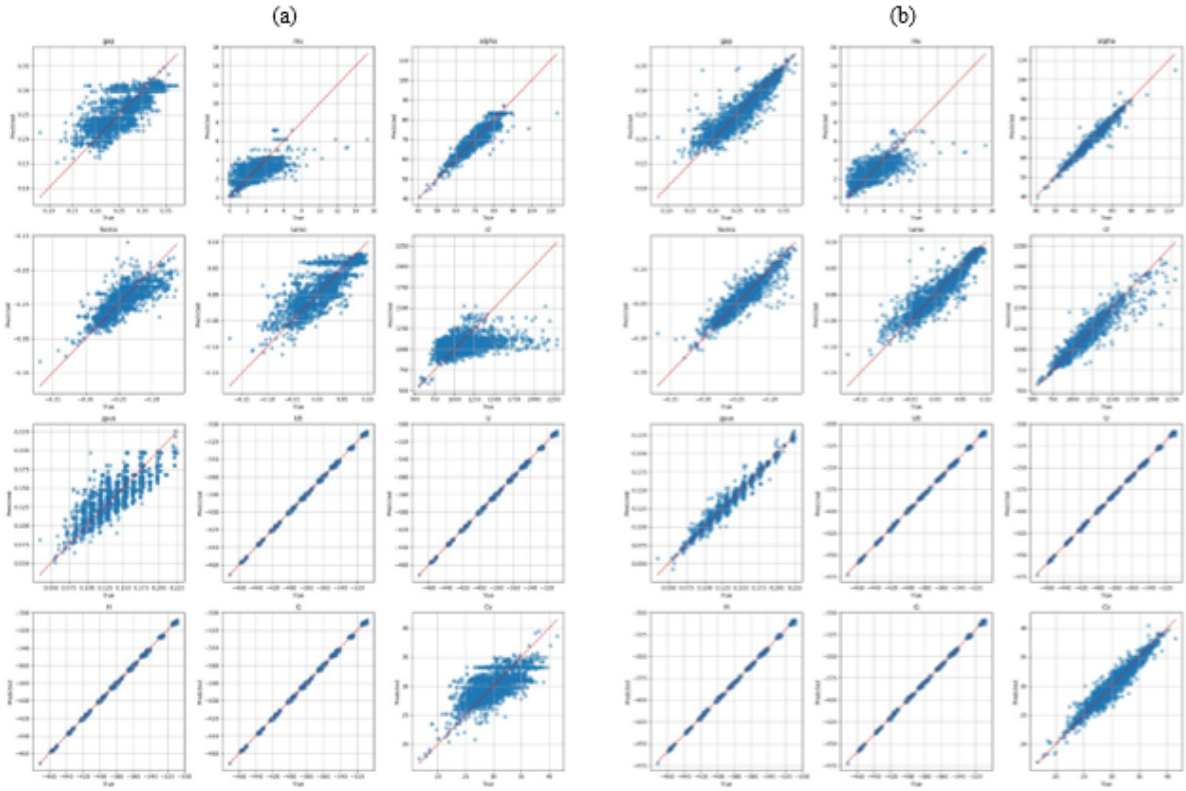


Figure 4: Parity plots (a) before and (b) after feature engineering for classical ML models

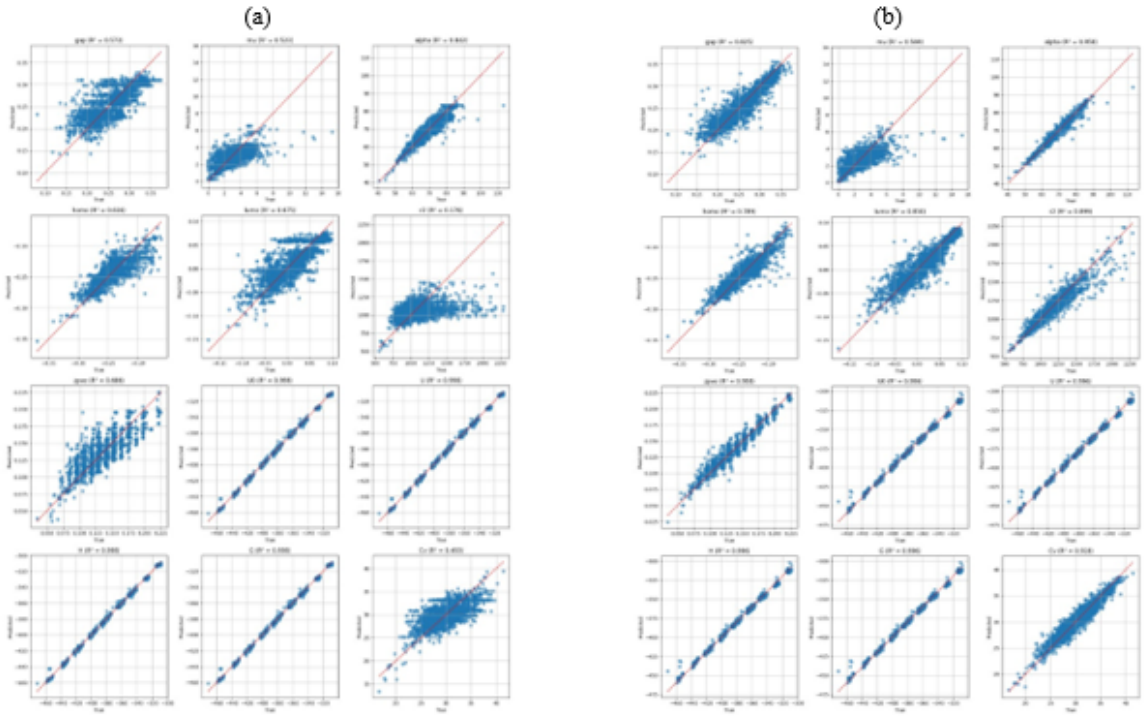


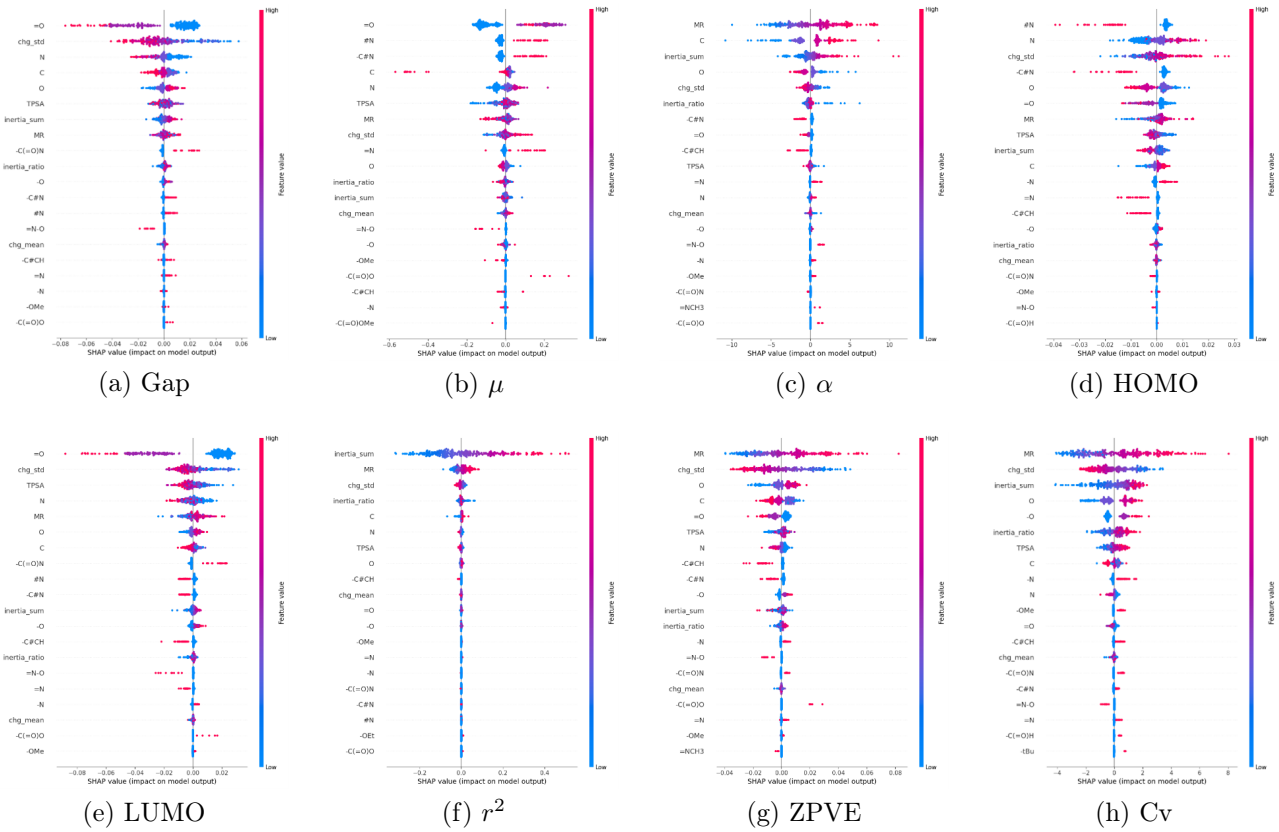
Figure 5: Parity plots (a) before and (b) after feature engineering for neural networks (see original report).

3.3. Comparison Between Classical Models and Neural Networks

After feature engineering, neural networks and classical machine learning show almost identical average performance. This is an important result because it demonstrates that, on structured molecular descriptors, sophisticated nonlinear deep models do not automatically outperform well-tuned classical algorithms.
Classical models retain two practical advantages. First, they are computationally lighter to optimize and interpret. Second, different regressors naturally specialize in different targets: Ridge is especially effective for highly linear thermodynamic quantities, while RF and HGBR perform well for nonlinear electronic and vibrational properties. Neural networks remain competitive, especially for C_v and r^2, but their gains over classical models are generally marginal.

3.4. Interpretability and Chemical Meaning

To interpret model predictions and assess the contribution of individual features, SHAP (SHapley Additive exPlanations) analysis was performed. The summary plots presented in Figure 6 (a) -(d) provide insights into the most influential descriptors for each target property.



(a) Gap (b) μ (c) α (d) HOMO

(e) LUMO (f) r^2 (g) ZPVE (h) Cv

Figure 6: SHAP summary plots for all target properties. Each panel highlights the most influential features for the corresponding prediction task.

For electronic properties such as the HOMO–LUMO gap and dipole moment, features related to charge distribution (e.g., chg_std) and the presence of electronegative functional groups (e.g., =O, nitriles) play a dominant role. This indicates that electronic asymmetry and local chemical environment strongly influence these properties.
For polarizability (α) and spatial extent (r^2), descriptors related to molecular size and shape, such as molar refractivity (MR) and inertia-based features, are the most significant contributors. These results align with physical expectations, as larger and more delocalized electron clouds lead to higher polarizability.
Thermodynamic and vibrational properties, such as ZPVE and Cv, are primarily influenced by atomic composition and molecular complexity. The importance of features such as carbon and oxygen counts, along with MR and charge variability, suggests that both structural and electronic factors contribute to vibrational behavior.
These findings validate the choice of descriptors: the model does not rely on arbitrary numerical artifacts but on variables that have an interpretable physical relation to the targets.

3.5. Why Dipole Moment Remains Difficult

Dipole moment is the most challenging property in the study. Even after feature engineering, its best R^2 remains around 0.58. This behavior is coherent with the underlying physics. Dipole moment is a vectorial quantity that depends sensitively on three-dimensional molecular geometry and subtle electronic asymmetries. Because the present framework is based on two-dimensional descriptors derived from SMILES and topological features, it cannot fully encode conformational orientation. The modest yet real improvement obtained from Gasteiger charge statistics and functional group descriptors indicates that electrostatic information helps, but it does not fully solve the problem.

4. Conclusion

This work presented a machine learning framework for predicting twelve quantum molecular properties from QM9 using functional groups, elemental composition, and physicochemical descriptors. The study confirms three main points.

First, feature engineering is the dominant factor in performance improvement. The transition from baseline descriptors to chemically informed features increases the average predictive quality from roughly 0.75 to roughly 0.91 in R^2 terms. Second, classical machine learning models remain highly competitive with neural networks when the input representation is rich and physically meaningful. Third, interpretability is preserved: SHAP analysis links model behavior to recognizable chemical concepts such as carbonyl content, charge heterogeneity, refractivity, and inertia.

The main remaining limitation concerns dipole moment, whose prediction is hindered by the absence of explicit three-dimensional geometry. Future work should therefore incorporate conformational descriptors, graph neural networks, or hybrid embeddings that combine learned representations with domain-based features. Even with this limitation, the present framework provides an accurate, scalable, and interpretable alternative to direct quantum-chemical computation for many molecular screening tasks.

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